



ANALOG CIRCUITS AND SENSOR SYSTEMS

PROJECT # 02

FILTER DESIGN PROJECT

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SPECIFICATIONS:

Amax[DB]	Amin[dB]	fp[kHz]	fs[kHz]	Filter Type	Gain	C Tolerance	R Tolerance
2	20	15	23	B	10	5%	1%

DESIGN STEPS:

In the design process, the first step is to calculate the epsilon (ϵ), which is determined by;

$$\epsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1} \quad \text{where } A_{max}=2$$

$$\epsilon=0.764$$

Once the ϵ is determined, the order of the filter is calculated using the expression; $A(\omega_s) \geq A_{min}$, where $A(\omega_s)$ is given as

$$A(\omega_s) = 10 \log [1 + \epsilon^2 (\omega_s / \omega_p)^{2N}]$$

Where, ω_s and ω_p are given as $2\pi f_s$ and $2\pi f_p$, respectively. From the above expressions (put different values of N and check), N is calculated to be,

$$N=7$$

Then, to design the filter, the Sallen-Key topology is chosen. For a 7th-order filter, 3 2nd-order and 1 1st-order stages are used. In this filter design procedure, Sallen & Key 'Smart' topology is chosen to implement these 2nd order filters. Since the values of Q are not very high, therefore, Sallen & Key topology is preferred. Using 'Smart' design process over the 'lazy' one because the 'lazy' procedure has high overshoot Now, poles position is determined using python:

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Poles for 7th-order Butterworth filter (fc = 15.59 kHz):
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Pole      Real Part (σ)      Imag Part (jω)      ω0 (rad/s)
-----
P1:  -21791.11      95473.10      97928.37      2.25
P2:  -61057.34      76563.48      97928.37      0.80
P3:  -88230.41      42489.53      97928.37      0.55
P4:  -97928.37      0.00      97928.37      0.50
P5:  -88230.41      42489.53      97928.37      0.55
P6:  -61057.34      76563.48      97928.37      0.80
P7:  -21791.11      95473.10      97928.37      2.25

Note: Complex poles come in conjugate pairs with identical ω0 and Q
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Additionally, the 2nd-order stages have a gain of 1 and the 1st-order is the gain stage with a 20dB gain.

Also, for low-pass Butterworth filter the transfer function is:

$$TF = \frac{1}{\{(s + 1)(s^2 + 0.4450s + 1)(s^2 + 1.2480s + 1)(s^2 + 1.8019s + 1)\}}$$

Quality factors of the 2nd-order terms of the transfer function are:

$$Q1 = 0.555$$

$$Q2 = 0.801$$

$$Q3 = 2.247$$

Then, to calculate the capacitors and resistors, these formulas are used:

$$\frac{C_2}{C_4} = 4Q^2$$

$$R = \frac{1}{\omega_0 \sqrt{C_2 C_4}}$$

However, the value of ω_0 is not clear and can't be equal to fp . So, we use the Butterworth filter magnitude formula to find the ω_0 .

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\frac{\omega}{\omega_0})^2}}$$

Then by $\omega = fp$ and $|H(jfp)| = 0.794$, we calculate ω_0 equal to 15.54KHz and therefore, the C1, C2 and R values can be calculated for each Q using the formulas above.

Calculations are done using the Python code attached to this file.

- Q = 0.555, C1 = 1nF, C2 = 0.81nF, R = 11.368 KΩ
- Q = 0.801, C1 = 1nF, C2 = 0.389nF, R = 16.4 KΩ
- Q = 2.247, C1 = 1nF, C2 = 46.5pF, R = 46 KΩ
- Gain stage, C = 100pF, R1 = 102.41 KΩ, R2 = 10.24 KΩ

1- Filter design using ideal op amp

.subckt ideal p m out params: Rin=1e9, Rout=1, A=1e9

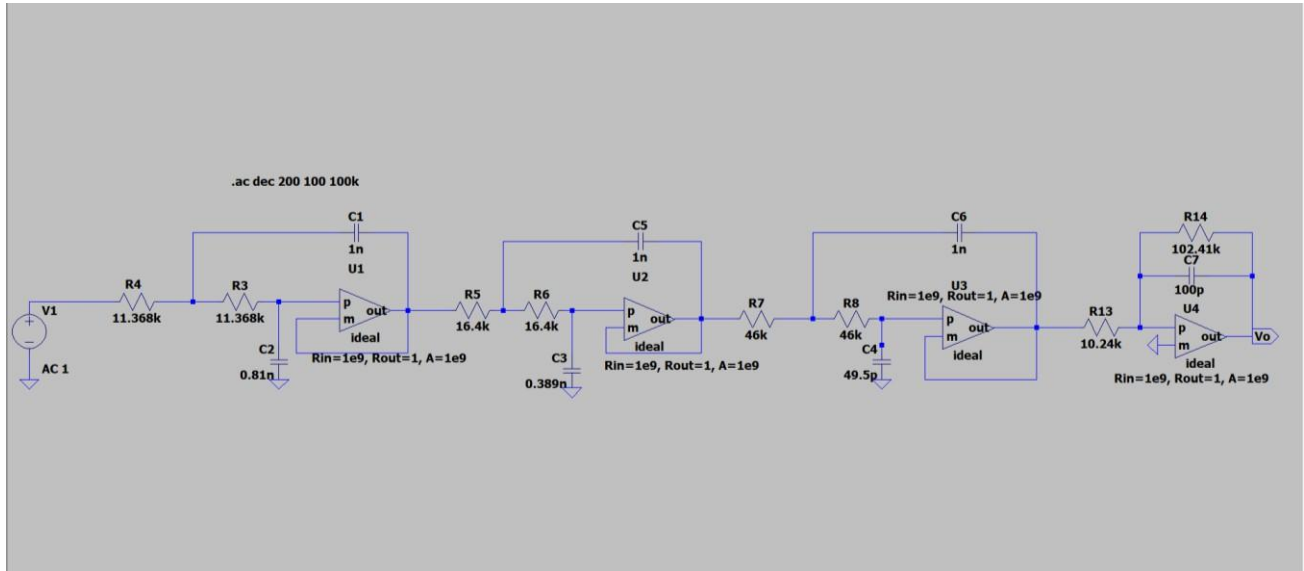
Rin p m {Rin}; input resistance

Eout out1 0 value={A*v(p,m)}; open-loop

voltage gain Rout out1 out {Rout} ; output

resistance

.ends ideal



2- Filter design using single-pole op amp

.subckt onepole p m out params: Rind=2e6, halfRinCm=1e9, Rout=75, TransC=0.19e-3, R1=1.323e9,

C1=100p Rind p m {Rind}

RinCm1 p 0 {halfRinCm}

RinCm2 m 0 {halfRinCm} ; input resistances

Gm int1 0

value={TransC*v(p,m)}

R1 int1 0 {R1}

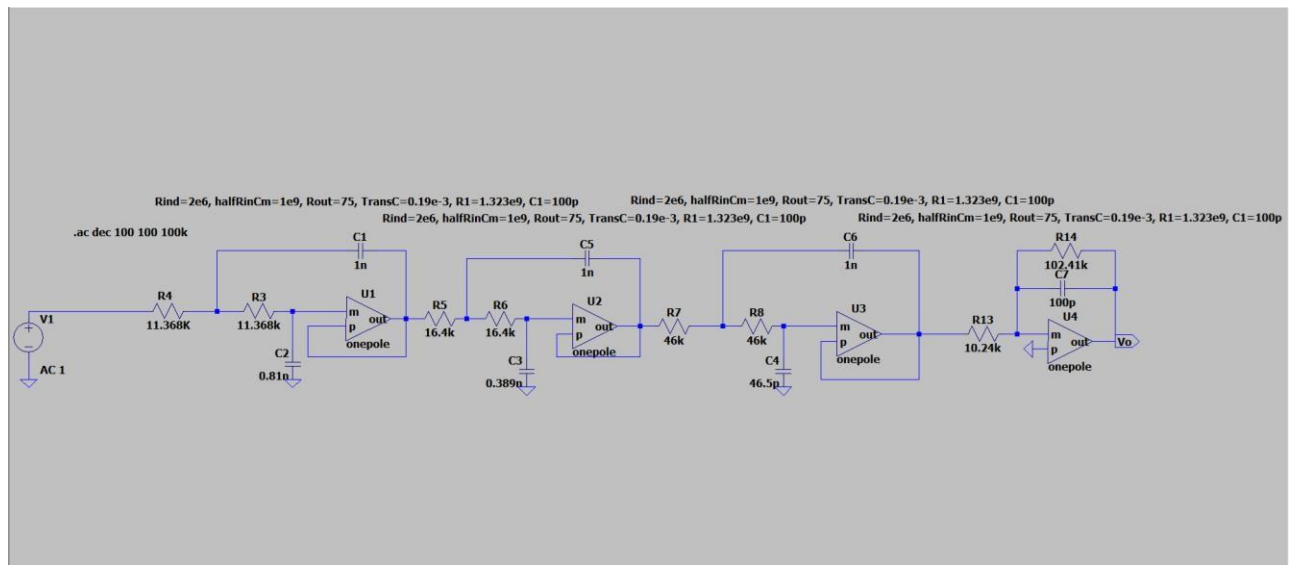
C1 int1 0 {C1} ; internal value

Eout out1 0 value={v(int1,0)} ; open-loop

voltage gain Rout out1 out {Rout} ; output

resistance

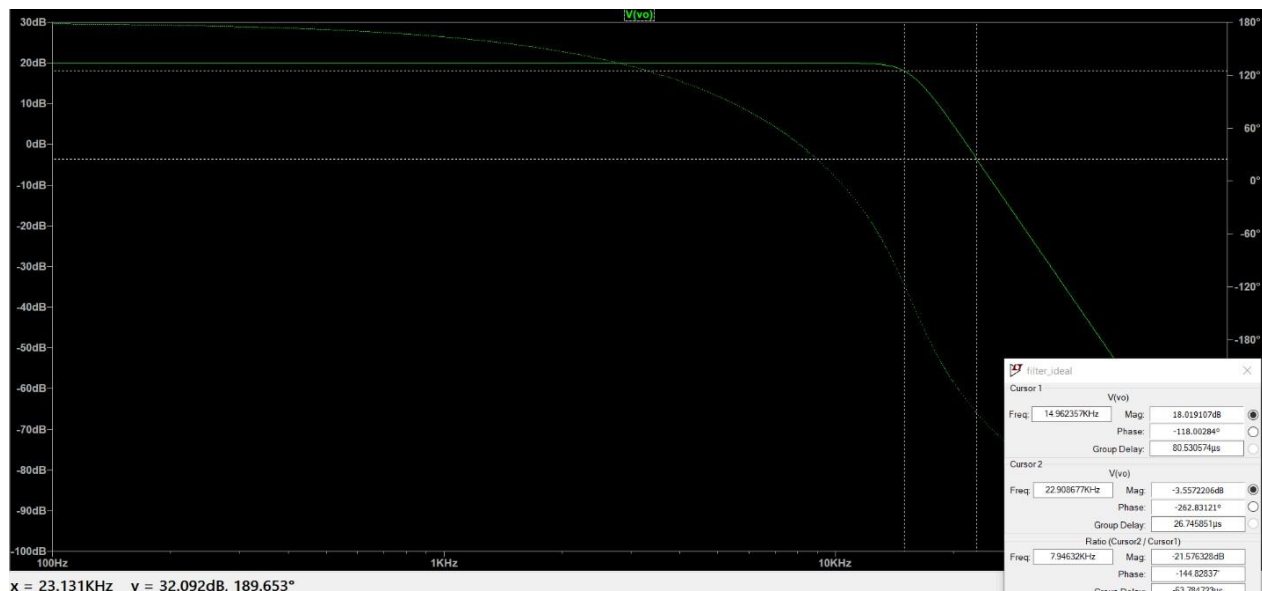
.ends onepole

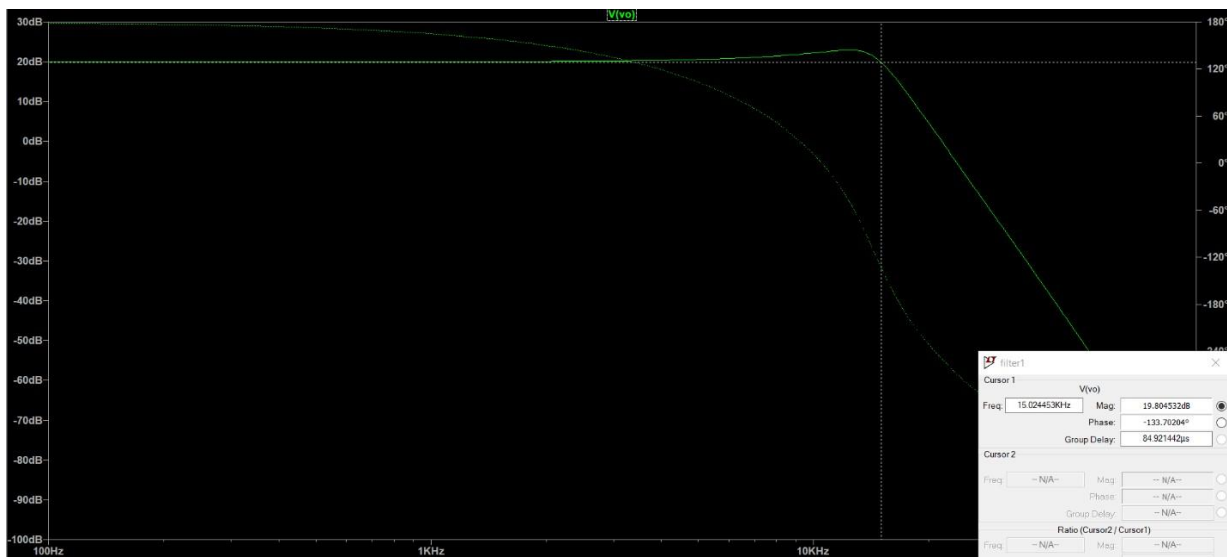


RESULTS

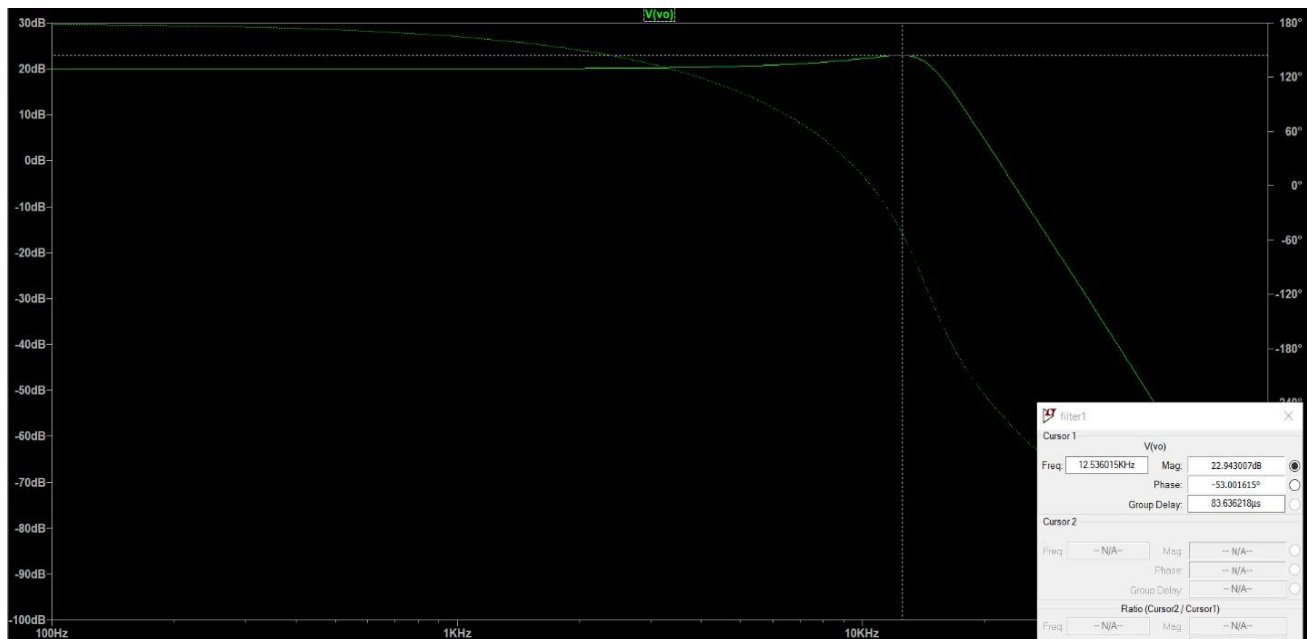
Bode plots of these filters are shown respectively

For the ideal op amp filter:





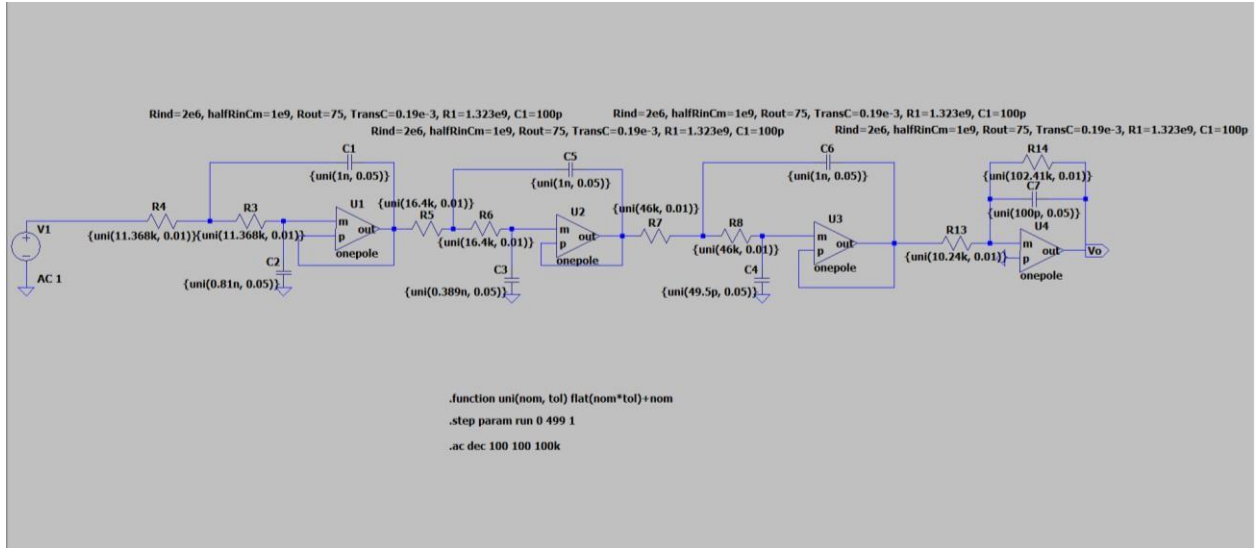
For the single-pole op amp filter:



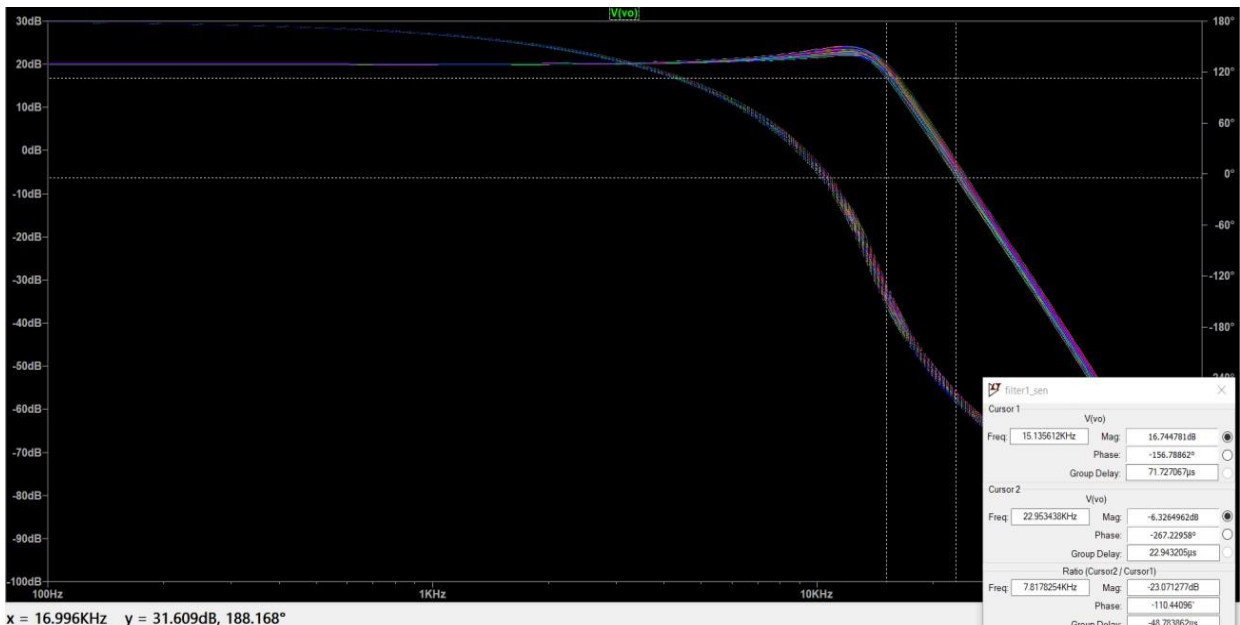
In this case, a 3dB ripple is created before the filter rolls off and it changes f_p value.

Sensitivity to parameter variations

The capacitors should be varied randomly with 5% and resistors with 1% tolerance, this is how it should be done.



And the result is:



It can be seen in the simulation results that the filter there are no overshoots since we used the S&K smart topology. Moreover, the response of the ideal filter from passband region to stopband region is steeper as we used the ideal Opamps within the circuit, whereas it is less steep in the case of the filter in which we used single-pole Opamp, which is more realistic